

```
import pandas as pd
import numpy as np
import io

#Read digit from digit.csv
from google.colab import files
uploaded = files.upload()
```

```
import pandas as pd
df = pd.read_csv(io.BytesIO(uploaded["digit.csv"]), header=None)
```

Double-click (or enter) to edit

df

Show First digit

[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0]							
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0]							
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0]							
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0]							
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	51	159
159	50	0	0	0	0	0	0	0	0	0]							
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	48	238	252
252	237	0	0	0	0	0	0	0	0	0]							
[	0	0	0	0	0	0	0	0	0	0	0	0	0	54	227	253	252
233	252	57	6	0	0	0	0	0	0	0]							
[	0	0	0	0	0	0	0	0	0	0	0	10	60	224	252	253	252
84	252	253	122	0	0	0	0	0	0	0]							
[	0	0	0	0	0	0	0	0	0	0	0	163	252	252	252	253	252
96	189	253	167	0	0	0	0	0	0	0]							
[	0	0	0	0	0	0	0	0	0	0	51	238	253	253	190	114	253
47	79	255	168	0	0	0	0	0	0	0]							
[	0	0	0	0	0	0	0	0	0	48	238	252	252	179	12	75	121
	0	0	253	243	50	0	0	0	0	0]							
[	0	0	0	0	0	0	0	0	38	165	253	233	208	84	0	0	0
	0	0	253	252	165	0	0	0	0	0]							
[	0	0	0	0	0	0	0	7	178	252	240	71	19	28	0	0	0
	0	0	253	252	195	0	0	0	0	0]							
[	0	0	0	0	0	0	0	57	252	252	63	0	0	0	0	0	0
	0	0	253	252	195	0	0	0	0	0]							
[	0	0	0	0	0	0	0	198	253	190	0	0	0	0	0	0	0
	0	0	255	253	196	0	0	0	0	0]							
[	0	0	0	0	0	0	76	246	252	112	0	0	0	0	0	0	0
	0	0	253	252	148	0	0	0	0	0]							
[	0	0	0	0	0	0	85	252	230	25	0	0	0	0	0	0	0
	7	135	253	186	12	0	0	0	0	0]							
[	0	0	0	0	0	0	85	252	223	0	0	0	0	0	0	0	0
131	252	225	71	0	0	0	0	0	0	0]							
[	0	0	0	0	0	0	85	252	145	0	0	0	0	0	0	0	48
252	173	0	0	0	0												

```
[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  
 0  0  0  0  0  0  0  0  0  0  0]  
[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  
 0  0  0  0  0  0  0  0  0  0  0]]
```

```
import matplotlib.pyplot as plt
```

Reshape 500x784 --> 10 x 50 x 784 digit x Number x pixel

```
digit = np.reshape(np.array(df.iloc[:, :-1]), [10, 50, 784]) # ignore the 1
```

Visualized a digit

Double-click (or enter) to edit

```
plt.imshow(np.reshape(digit[2, 15, :], [28, 28]), cmap='gray')  
plt.show()
```

Try Visualize one of digit 3

```
plt.imshow(np.reshape(digit[3, 15, :], [28, 28]), cmap='gray')
```

```
print(digit[0,:,:].shape)
```

```
(50, 784)
```

✓ Binarize the digit

```
bin_digit=1*(digit>128)
```

✓ Create train data (80%/20%)

```
number0=np.mean(bin_digit[0,:40,:],axis=0)# P(D|0)  
number1=np.mean(bin_digit[1,:40,:],axis=0)# P(D|1)  
  
print(number0.shape)
```

```
(784,)
```

```
print(number0)
```

[0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.025	0.	0.	0.025	0.025	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.025	0.025	0.125	0.275	0.2	0.275	0.35	0.35	0.25	0.075
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.05	0.075	0.225	0.35	0.5	0.675	0.75
0.7	0.65	0.5	0.35	0.175	0.05	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.05	0.125	0.3	0.25
0.5	0.6	0.8	0.875	0.825	0.7	0.75	0.475	0.35	0.15	0.025	0.025
0.025	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.05
0.1	0.2	0.275	0.475	0.575	0.825	0.825	0.775	0.8	0.875	0.8	0.75
0.55	0.25	0.05	0.025	0.025	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.025	0.075	0.225	0.275	0.45	0.6	0.675	0.725	0.65	0.575
0.575	0.725	0.75	0.825	0.7	0.425	0.125	0.05	0.025	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.075	0.125	0.275	0.4	0.575	0.8
0.65	0.625	0.5	0.375	0.4	0.45	0.55	0.75	0.775	0.55	0.3	0.075
0.05	0.	0.	0.	0.	0.	0.	0.	0.	0.025	0.125	0.225
0.325	0.525	0.7	0.6	0.575	0.525	0.4	0.25	0.2	0.25	0.35	0.55
0.775	0.625	0.425	0.125	0.05	0.	0.	0.	0.	0.	0.	0.
0.	0.075	0.2	0.3	0.4	0.6	0.625	0.6	0.475	0.3	0.2	0.1
0.1	0.15	0.275	0.425	0.625	0.575	0.525	0.225	0.1	0.	0.	0.
0.	0.	0.	0.	0.	0.1	0.275	0.35	0.55	0.675	0.625	0.525
0.325	0.225	0.075	0.05	0.025	0.1	0.275	0.35	0.625	0.575	0.5	0.275
0.125	0.	0.	0.	0.	0.	0.	0.	0.	0.15	0.325	0.425
0.575	0.65	0.525	0.375	0.225	0.05	0.025	0.025	0.025	0.05	0.275	0.325
0.625	0.575	0.55	0.3	0.125	0.	0.	0.	0.	0.	0.	0.
0.	0.175	0.375	0.575	0.625	0.625	0.4	0.275	0.1	0.025	0.	0.
0.	0.075	0.2	0.35	0.575	0.575	0.45	0.3	0.15	0.	0.	0.
0.	0.	0.	0.	0.	0.2	0.425	0.575	0.75	0.55	0.325	0.125
0.025	0.	0.	0.	0.	0.05	0.175	0.425	0.625	0.55	0.425	0.275
0.125	0.	0.	0.	0.	0.	0.	0.	0.	0.25	0.45	0.65
0.7	0.5	0.25	0.075	0.025	0.	0.	0.	0.	0.025	0.275	0.5
0.625	0.575	0.325	0.225	0.1	0.	0.	0.	0.	0.	0.	0.
0.025	0.3	0.525	0.675	0.675	0.55	0.2	0.1	0.05	0.	0.	0.
0.05	0.15	0.4	0.575	0.675	0.525	0.275	0.225	0.1	0.	0.	0.
0.	0.	0.	0.	0.	0.25	0.5	0.75	0.725	0.525	0.25	0.075
0.05	0.	0.025	0.075	0.125	0.4	0.6	0.65	0.5	0.325	0.25	0.15
0.05	0.	0.	0.	0.	0.	0.	0.	0.	0.175	0.45	0.75
0.85	0.65	0.4	0.2	0.05	0.075	0.15	0.225	0.45	0.55	0.7	0.625
0.4	0.35	0.175	0.125	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.075	0.35	0.625	0.85	0.8	0.65	0.45	0.375	0.475	0.45	0.525
0.65	0.725	0.575	0.425	0.325	0.225	0.15	0.025	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.025	0.15	0.5	0.725	0.9	0.825	0.725
0.7	0.675	0.7	0.775	0.775	0.675	0.5	0.275	0.25	0.15	0.05	0.
0.	0.	0.	0.	0.	0.	0.	0.	0.	0.025	0.05	0.225
0.55	0.75	0.85	0.9	0.825	0.875	0.875	0.775	0.65	0.45	0.3	0.225
0.125	0.05	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.05	0.05	0.2	0.425	0.6	0.725	0.75	0.675	0.65	0.4

0.25	0.2	0.175	0.05	0.025	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.05	0.05	0.05	0.1	0.175
0.25	0.175	0.2	0.15	0.075	0.05	0.05	0.	0.	0.	0.	0.

```
plt.imshow(np.reshape(number0,[28,28]),cmap='gray')
```

```
plt.imshow(np.reshape(number1,[28,28]),cmap='gray')
```

```
print(np.reshape(number1,[28,28]))
```

```
[[0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.  0.  0.025 0.025 0.025 0.  0.  0.  0.  0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.  0.025 0.025 0.  0.
  0.125 0.125 0.125 0.2  0.25 0.225 0.125 0.125 0.05 0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.025 0.025 0.025 0.  0.0
  0.2  0.25 0.25 0.35 0.425 0.425 0.175 0.225 0.1  0.025 0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.025 0.025 0.025 0.025 0.025 0.0
  0.175 0.25 0.35 0.4  0.525 0.45 0.3  0.25 0.1  0.025 0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.025 0.025 0.025 0.05 0.0
  0.175 0.275 0.45 0.425 0.575 0.375 0.325 0.225 0.1  0.025 0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.  0.025 0.05 0.05 0.0
  0.175 0.3  0.525 0.6  0.575 0.425 0.325 0.15 0.05 0.025 0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.  0.025 0.05 0.075 0.0
  0.15 0.3  0.6  0.675 0.65 0.4  0.25 0.1  0.025 0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.  0.025 0.05 0.05 0.0
  0.125 0.325 0.75 0.8  0.65 0.3  0.125 0.05 0.  0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.025 0.025 0.05 0.075 0.0
  0.1  0.475 0.85 0.85 0.525 0.25 0.1  0.  0.  0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.025 0.05 0.05 0.025 0.025 0.0
  0.125 0.55 0.925 0.85 0.475 0.175 0.025 0.  0.  0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.025 0.025 0.  0.  0.  0.
  0.125 0.75 0.925 0.85 0.4  0.1  0.  0.  0.  0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.
  0.175 0.825 0.95 0.9  0.275 0.075 0.  0.  0.  0.  0.
  0.  0.  0.  0.  ]
 [0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.0
  0.325 0.925 0.975 0.625 0.125 0.025 0.  0.  0.  0.  0.]
```

```

0.    0.    0.    0.    ]
[0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.1
0.45  0.9   0.95  0.475 0.15  0.025 0.025 0.    0.    0.    0.    0.
0.    0.    0.    0.    ]
[0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.05  0.2
0.5   0.875 0.825 0.425 0.1   0.025 0.05  0.025 0.025 0.    0.    0.
0.    0.    0.    0.    ]
[0.    0.    0.    0.    0.    0.    0.    0.    0.    0.    0.15  0.3
0.6   0.85  0.675 0.375 0.15  0.05  0.05  0.05  0.05 0.    0.    0.
0.    0.    0.    0.    ]
[0.    0.    0.    0.    0.    0.    0.    0.    0.    0.075 0.25  0.4

```

```
(1-digit[0,40:,:])*(1-number0)
```

```

array([[1., 1., 1., ..., 1., 1., 1.],
       [1., 1., 1., ..., 1., 1., 1.],
       [1., 1., 1., ..., 1., 1., 1.],
       ...,
       [1., 1., 1., ..., 1., 1., 1.],
       [1., 1., 1., ..., 1., 1., 1.],
       [1., 1., 1., ..., 1., 1., 1.]])

```

```

# p(0|number) = p(digit|0) p(0) / (p(0|digit) p(0)+ p(1|digit) p(1))
p0 = np.prod(bin_digit[0,40:,:]*number0+(1-bin_digit[0,40:,:])*(1-numk
p1 = np.prod(bin_digit[0,40:,:]*number1+(1-bin_digit[0,40:,:])*(1-numk
# this is using input as digit 0 * probability of being 0 for each bit
prob0=p0/(p0+p1)
prob1=p1/(p0+p1)

```

```

/tmp/ipython-input-2223695456.py:5: RuntimeWarning: invalid value encou
prob0=p0/(p0+p1)
/tmp/ipython-input-2223695456.py:6: RuntimeWarning: invalid value encou
prob1=p1/(p0+p1)

```

```
print(np.array([prob0 ,prob1]).T)
```

```

[[ 1.  0.]
 [ 1.  0.]
 [nan nan]
 [nan nan]
 [ 1.  0.]
 [ 1.  0.]
 [nan nan]
 [nan nan]
 [nan nan]
 [ 1.  0.]]

```

Start coding or [generate](#) with AI.


```
p0 = np.prod(bin_digit[1,40:,:]*number0+(1-bin_digit[1,40:,:])*(1-numk
p1 = np.prod(bin_digit[1,40:,:]*number1+(1-bin_digit[1,40:,:])*(1-numk

prob0=p0/(p0+p1)
prob1=p1/(p0+p1)
```

```
[[0. 1.]
 [0. 1.]
 [0. 1.]
 [0. 1.]
 [0. 1.]
 [0. 1.]
 [0. 1.]
 [0. 1.]
 [0. 1.]
 [0. 1.]]
```

```
number=[]
for i in range(10):
    number.append(np.mean(bin_digit[i,:40,:],axis=0))
```

```
[4.44765414e-060 1.48737941e-072 1.41769614e-100 1.89307428e-080  
4.06044816e-081 2.81165087e-083 3.53461067e-102 7.78183500e-113  
2.47855612e-081 4.01152064e-074 4.90811169e-072 5.29905002e-079  
2.56435269e-064 2.87046460e-113 1.95223110e-079 2.66023003e-067  
7.50068114e-080 2.06794574e-101 1.52431864e-089 4.98477615e-075  
2.21269763e-064 3.19671219e-093 1.88642140e-090 4.72663100e-075  
5.22717685e-088 1.68824839e-090 3.37514541e-092 6.51703901e-077  
6.84830312e-118 2.41208614e-093 6.40960294e-101 5.97919143e-082  
1.00325623e-069 9.80879096e-070 9.43390067e-062 1.56235752e-065  
1.09591521e-057 5.25248276e-062 9.76002913e-079 3.46290381e-071]  
[0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.  
0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0. 0.]  
[2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2. 2.
```